



GIA - IAA

Introducció a

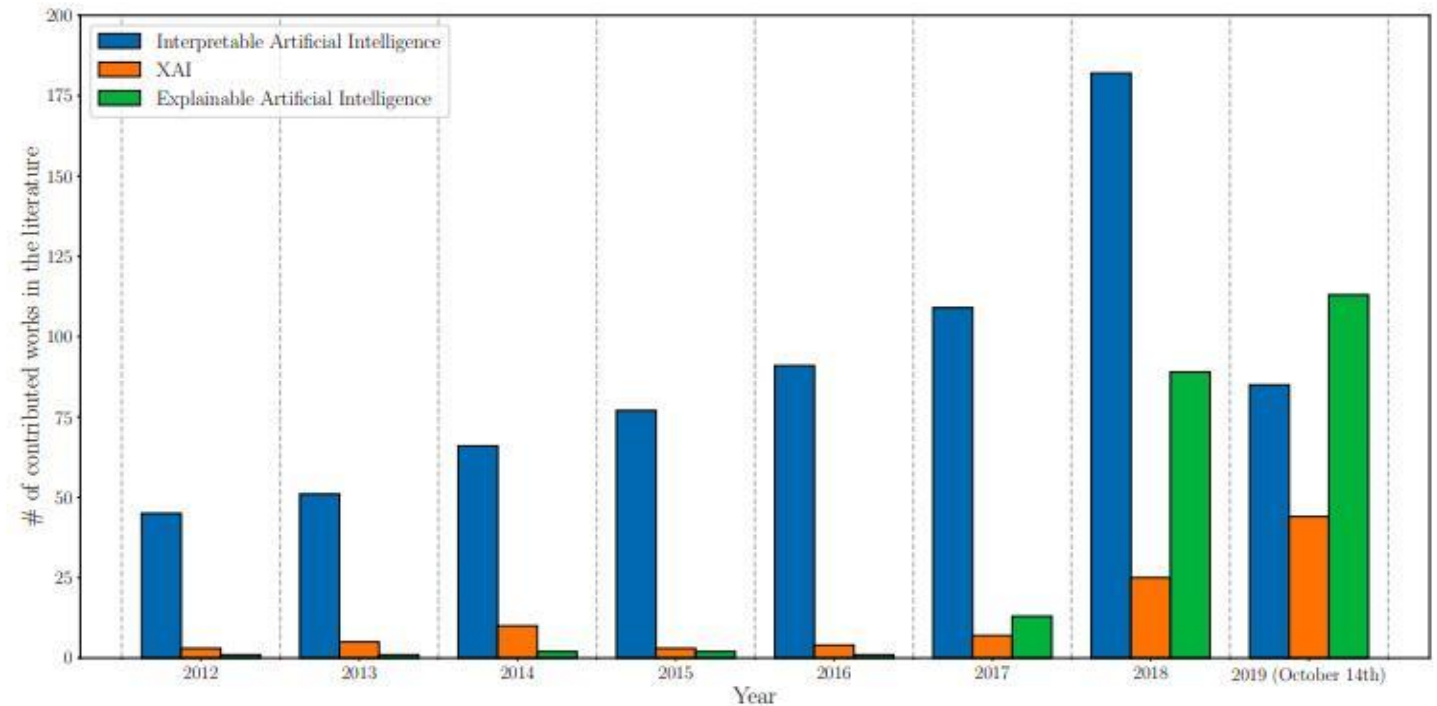
l'Aprenentatge Automàtic

11. Lab: XAI (eXplainable Artificial Intelligence)

Jordi Luque
jordi.luque.serrano@upc.edu

Explainable Machine Learning (XAI)

- ❖ Interpretability is the degree to which a human can understand the cause of a decision – [Miller, Tim](#)



What is Interpretability?

- ❖ Interpretability is the degree to which a human can understand the cause of a decision. And in the artificial intelligence domain, it means it is the degree to which a person can understand the how and why of an algorithm and its predictions.

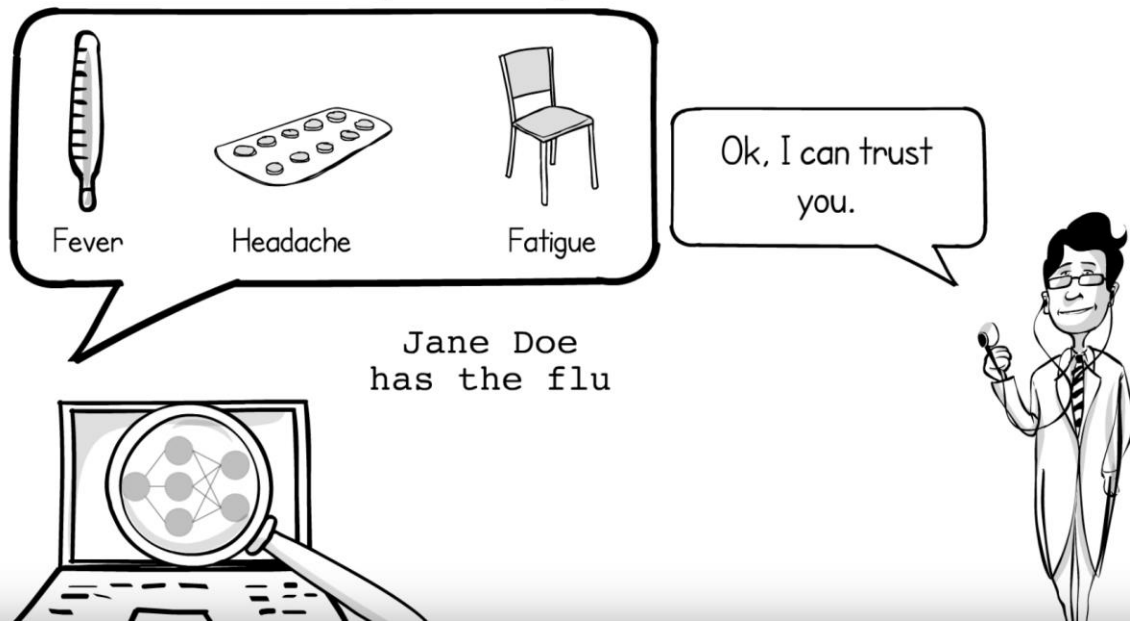
Why XAI?

- ❖ What if you could understand why any classifier is making a prediction?
 - Choose between competing models
 - Detect and improve untrustworthy models
 - Get insights into the model

Why XAI?

- ❖ What if you could understand why any classifier is making a prediction?

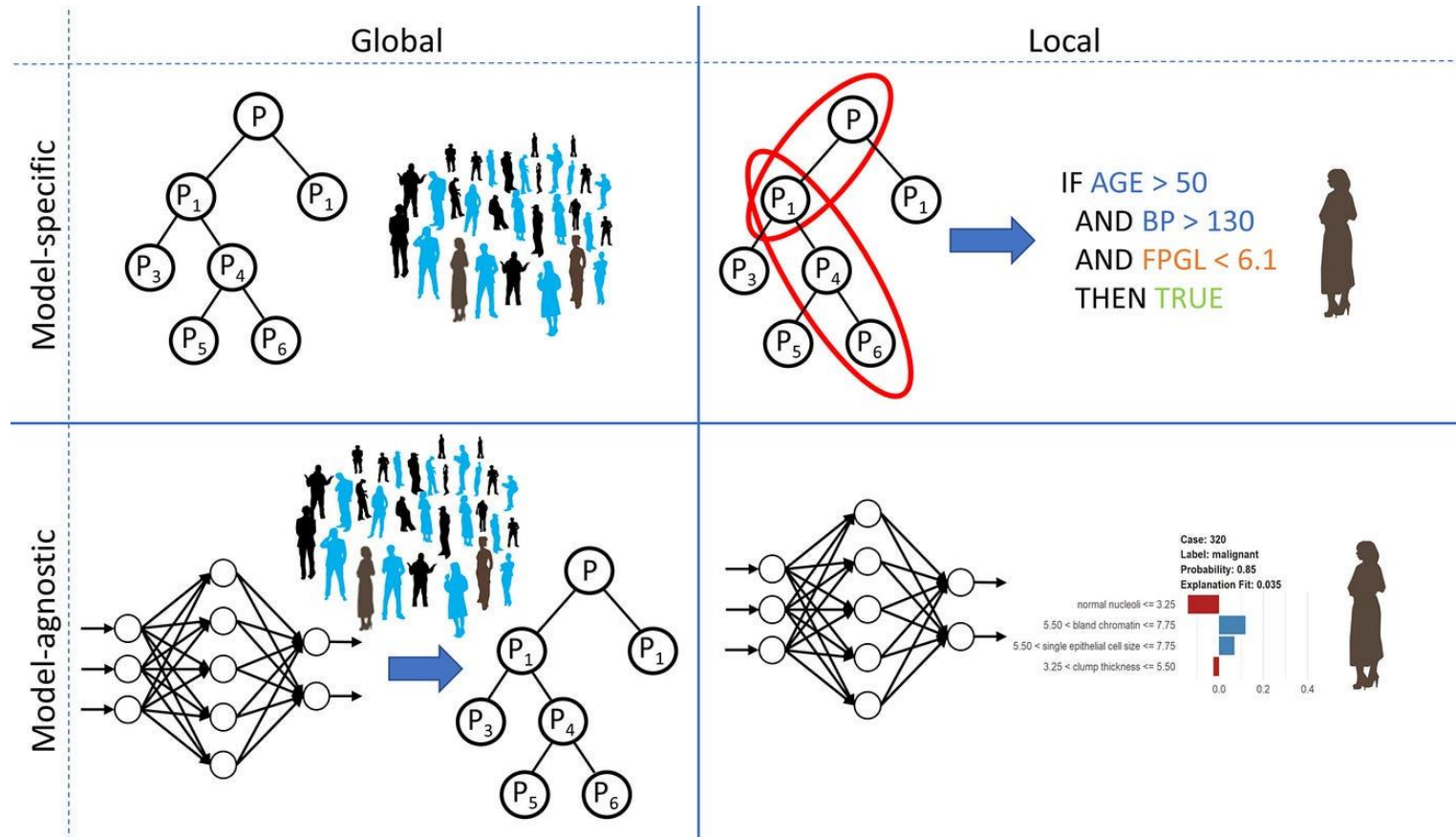
It's easier to trust a prediction if you understand the reasons for it.



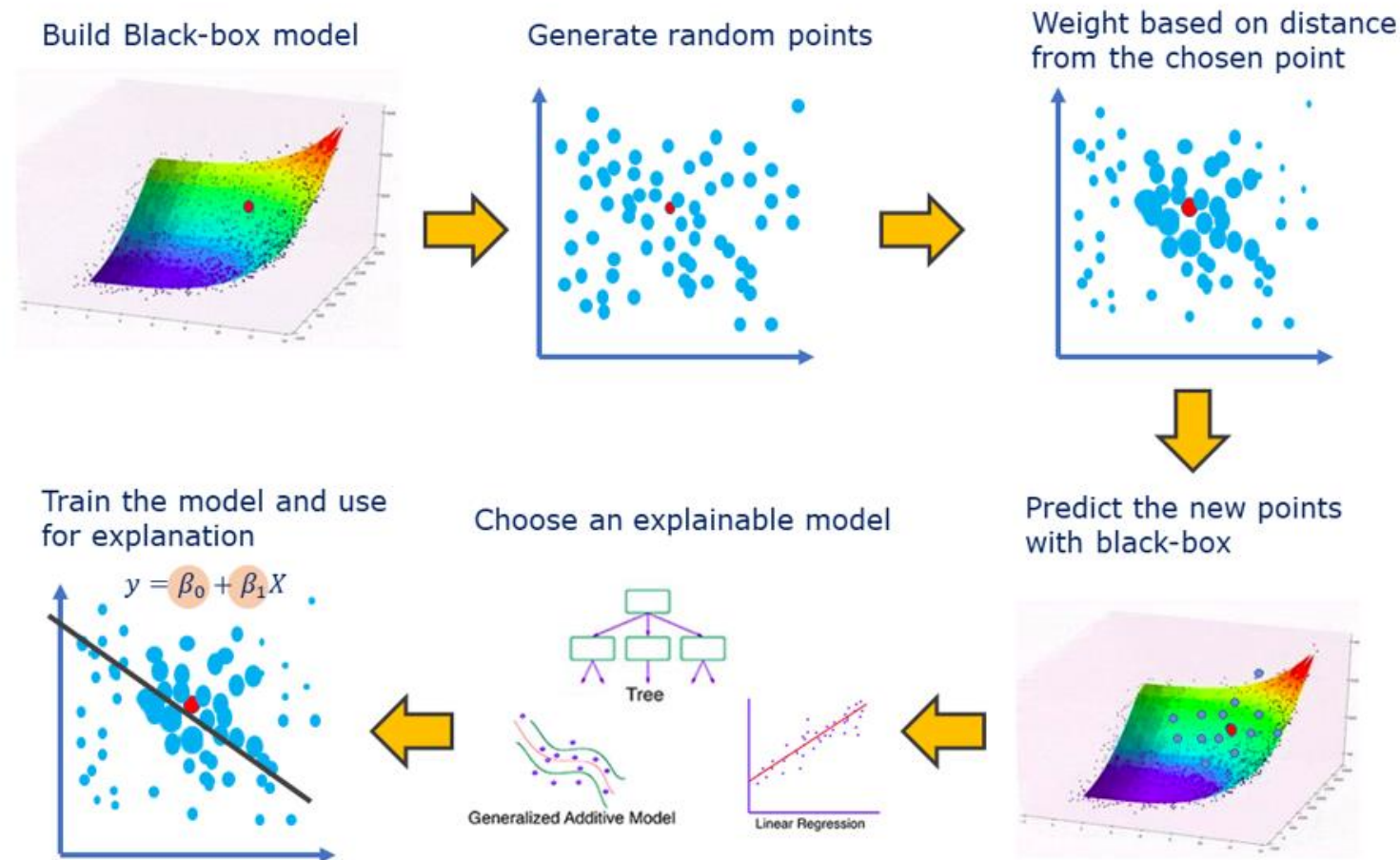
Or figuring out when you shouldn't trust a model



Model-specific/agnostic and Global/Local Explanations

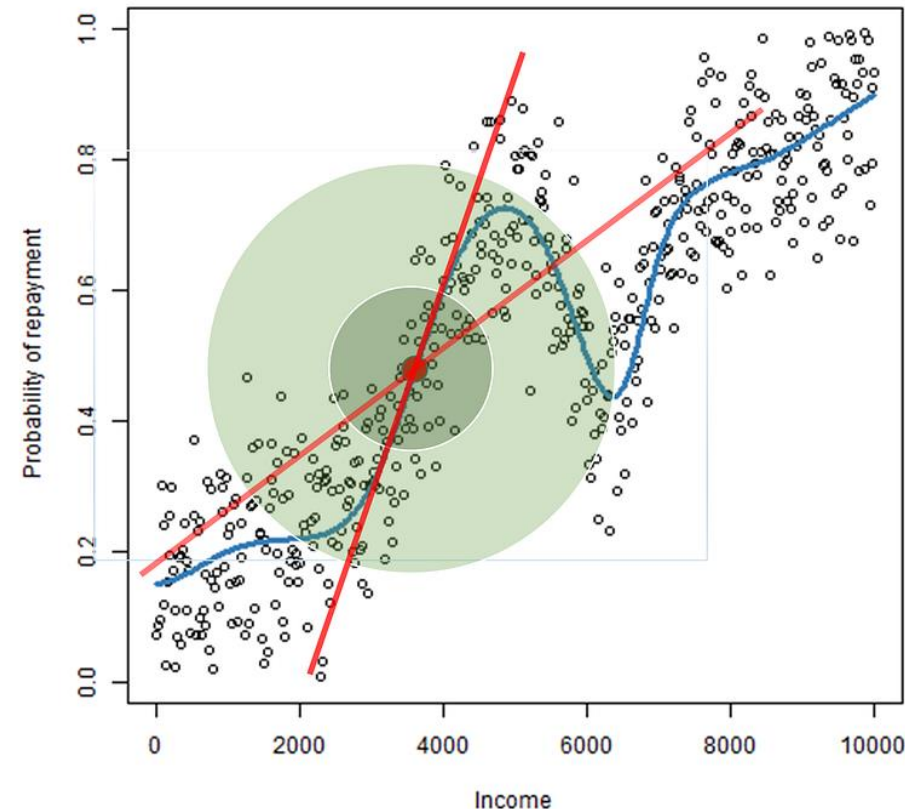


LIME (Local Interpretable Model-agnostic Explanations)



LIME (Local Interpretable Model-agnostic Explanations)

- ❖ The generation step is an open issue!
- ❖ Why don't just generate only points close to the reference?
- ❖ How to choose the correct value for the kernel width?

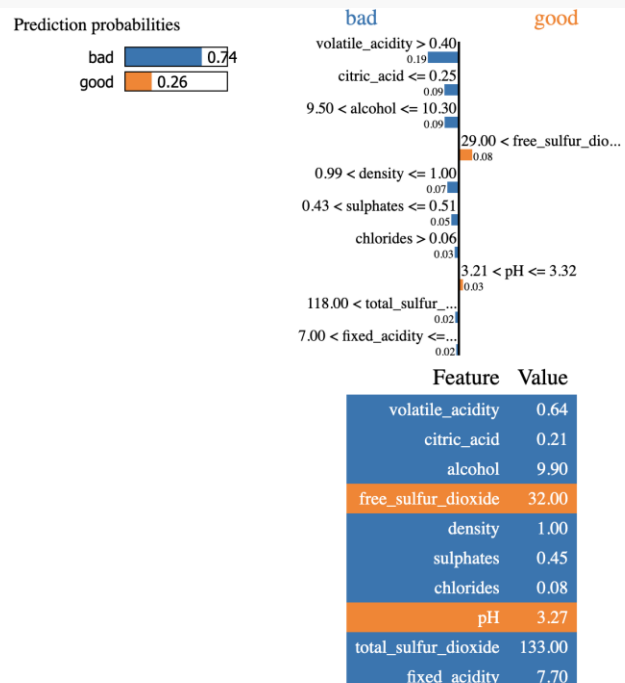


$$\xi = \arg \min_{g \in \mathcal{G}} L(f, g, \pi_{x'}) + \Omega(g).$$

LIME (Local Interpretable Model-agnostic Explanations)

```
1 lime_explainer = lime_tabular.LimeTabularExplainer(  
2     training_data=np.array(X_train),  
3     feature_names=X_train.columns,  
4     class_names=['bad', 'good'],  
5     mode='classification'  
6 )  
7  
8  
9 lime_exp = lime_explainer.explain_instance(  
10     data_row=test_1,  
11     predict_fn=lgb_opt.predict_proba  
12 )  
13 lime_exp.show_in_notebook(show_table=True)
```

✓ 0.1s



[Docs](#) » Local Interpretable Model-Agnostic Explanations (lime)

[Edit on GitHub](#)

Local Interpretable Model-Agnostic Explanations (lime)

In this page, you can find the Python API reference for the lime package (local interpretable model-agnostic explanations). For tutorials and more information, visit [the github page](#).

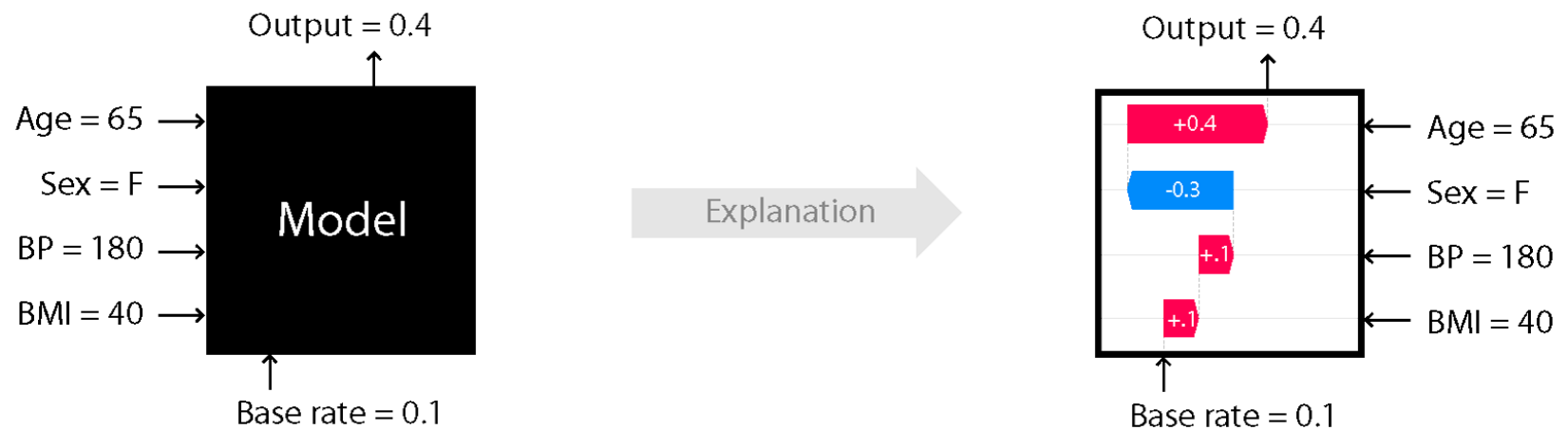
- lime package
 - Subpackages
 - Submodules
 - lime.discretize module
 - lime.exceptions module
 - lime.explanation module
 - lime.lime_base module
 - lime.lime_image module
 - lime.lime_tabular module
 - lime.lime_text module
 - lime.submodular_pick module
 - Module contents

Indices and tables

- Index
- Module Index
- Search Page

SHAP (SHapley Additive exPlanations)

- ❖ SHAP is a game theoretic approach to explain the output of any machine learning model. It connects optimal credit allocation with local explanations using the classic Shapley values from game theory



SHAP (SHapley Additive exPlanations)

- ❖ Shapley values represent the amount of credit or blame each feature holds for the model output
- ❖ One of the fundamental properties of Shapley values is that they always sum up to the difference between the game outcome when all players are present and the game outcome when no players are present. For machine learning models this means that SHAP values of all the input features will always sum up to the difference between baseline (expected) model output and the current model output for the prediction being explained.

SHAP (SHapley Additive exPlanations)

$$v(c) = \begin{cases} 80, & \text{if } c = \{A\} \\ 56, & \text{if } c = \{B\} \\ 70, & \text{if } c = \{C\} \\ 80, & \text{if } c = \{A, B\} \\ 85, & \text{if } c = \{A, C\} \\ 72, & \text{if } c = \{B, C\} \\ 90, & \text{if } c = \{A, B, C\} \end{cases}$$

π	δ_{π}^G
(A, B, C)	$(80, 0, 10)$
(A, C, B)	$(80, 5, 5)$
(B, A, C)	$(24, 56, 10)$
(B, C, A)	$(18, 56, 16)$
(C, A, B)	$(15, 5, 70)$
(C, B, A)	$(18, 2, 70)$

$$\xi = \arg \min_{g \in \mathcal{G}} L(f, g, \pi_{x'}) + \Omega(g).$$

- ❖ Alice, Bob, and Celine share a meal. The bill came to be 90, what they each owe?
- ❖ The marginal contribution of A = $(80+80+56+16+5+70)/6 = 51.17$. And if we calculate the expected marginal contributions of each of the players and add them together, we will get 90- which is the total payoff if all three ate together